# LA GRANDEE INTERNATIONAL COLLEGE

## Simalchaur, Pokhara Nepal

A Project Proposal

On

## "Rock, Paper, Scissor"

**Submitted to:**

Bachelor of Computer Application (BCA) Program

In partial fulfilment of the requirements for the degree of BCA under

Pokhara University

**Submitted by:**


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# 1. Introduction

Rock, Paper, Scissors is one of the most popular and oldest hand games played throughout the world. The game is simple, entertaining and commonly used for decision making purposes. Traditionally , the game is played between two players where each player chooses whether rock, paper, or scissors .In this computerized version, the user plays against the computer ,and the winner is determined according to the rules: rock defeats scissors, scissors defeat paper, and paper defeats rocks.

This project focuses on developing an advanced Rock Paper Scissors game using C programming language with several advanced features such as login system, registration system, CRUD operations, score management, multiplayer gameplay, account deletion, tournament mode, leaderboard system, match history, and file handling .The project is designed to simulate a real-world gaming system where users can create accounts, store scores permanently, compete with other players, and manage their data efficiently .

The system will allow users to create accounts and securely log in using usernames. User data is stored permanently using file handling techniques .Once logged in, players can play against the computer or compete with another player in multiplayer mode. Scores are recorded and stored in files so that player performance can be tracked even after the program is closed.

The implementation of CRUD operation is one of the most important highlights of the project. CRUD stands for Create, Read, Update, and Delete. In this project, users can create new accounts, read player details and scores, update records automatically through gameplay, and delete accounts permanently.

File handling is another important component of the project .Since the C language does not include a built-in database management system, text files are used to store player records, scores, and match history. The multiplayer feature increases interaction and competitiveness between users. Instead of limiting gameplay to computer-based interaction only, the system allows two users to compete directly against each other. This improves our experience and makes the game more engaging.

This project also includes additional innovative feature such as tournament mode and streak bonus system. Tournament mode allows users to compete in multiple rounds continuously, while the streak bonus system reward users for consecutive wins. These features increase excitement and provide better gaming experience.

The project demonstrates the practical application of various programming concepts including loops, conditional statements, arrays, structures, functions, file handling and user authentication which helps beginner programmers understand how multiple concepts can be combined together to create a complete software application.

Overall, the advanced Rock Paper Scissors game is an educational and interactive project that combines entertainment with software development knowledge. It provides practical understanding of game logic, user management systems, and permanent data storage through file handling.

## 2. Objective

The main objective of this project is to develop a complete Rock Paper Scissors gaming system using C programming language. Here are some specific objectives listed below:

- To implement multiple gameplay modes including player versus computer and multiplayer functionality.
- To use file handling for storing player records, maintaining match history, and displaying scoreboards.
- Dashboard that provides easy access to gameplay, match history, leaderboard and user options.

# 3. Study of Previous System

The previous Rock Paper Scissors system was a simple game developed to provide basic entertainment for users. The system allowed a player to play against the computer by selecting one of three options: rock, paper, or scissors. The computer generated its choice randomly, and the winner was decided according to the game rules. The system was easy to use and helped user understand the basic concept of game programming.

**Features of the Previous System**

- Basic single-player gameplay.
- Random computer-generated moves.
- Simple game logic implementation.
- Basic score display during gameplay.
- Simple user interface.

**Limitations of Previous System**

- The game was limited to single-player mode only.
- There was no user registration and login system.
- Scores and match history were not stored permanently.
- No multiplayer game feature was available.
- The game lacked a scoreboard and ranking system.
- CRUD operations were not implemented properly.
- The interface was less interactive and user-friendly.
- Data handling and file management were inefficient.
- The game had limited features and low user engagement.

# 4. Current Problem and Proposed Solution

### Current Problems

- There Traditional games don’t save player records permanently.
- Users cannot create personal accounts.
- There is no secure login system.
- Players cannot track scores and ranking.
- Multiplayer mode is missing in basic systems.
- Existing systems lack CRUD operations.
- Match history cannot be stored.

### Proposed Solutions

To solve the above problems, an advanced Rock Paper Scissors game will be developed using C programming language.

The proposed system will include:

- User registration and login system.
- File handling for permanent data storage.
- Multiplayer gameplay mode.
- Score management system.
- Match history feature.
- Leaderboard system.
- CRUD operations.
- Streak bonus feature.

The proposed system will improve user interaction, provide better learning opportunities, and demonstrate real-world software development concepts.

# 5. High Level Requirements

High level requirements describe the major features and functionalities that the proposed Rock Paper Scissors game system should provide. These requirements define what the system is expected to achieve without going into technical details.

**5.1 Functional Requirements:**

Functional requirements describes the functions and services provided by the system.

- The system will allow users to register accounts.
- The system will allow users to log in.
- The system will verify usernames and passwords.
- The system will allow players to play both single-player and multiplayer modes.
- The system will maintain scores.
- The system will display leaderboard rankings.
- The system will save match history.
- The system will allow account deletion.
- The system will store data using file handling.
- The system will support tournament mode.
- The system will provide streak bonus rewards.

**5.2 Non-Functional Requirements:**

Non-functional requirement describe the quality and performance of the system.

- The system will be user-friendly.
- The system will execute efficiently.
- The system will store data securely.
- The system will be easy to maintain.
- The system will provide fast response time.
- The system will be reliable.
- The system will handle file operations properly.
- The system will be simple for beginners to use.
- The system will provide accurate results.

### 5.3 Requirement Prioritization Table

| S.N | Requirement | Priority Level | Description |
|---|---|---|---|
| 1 | User registration and login | High | Required for identifying and managing users. |
| 2 | Single-Player mode | High | Core gameplay feature against computer |
| 3 | Multiplayer mode | High | Enhances user interaction and engagement |
| 4 | Player choice selection (rock/paper/scissors) | High | Basic game functionality |
| 5 | Winner determination | High | Core game logic |
| 6 | Score display | High | Shows result after each round |
| 7 | File handling (store scores and history) | High | Ensures permanent data storage |
| 8 | CRUD operations | High | Required for managing user data |
| 9 | Scoreboard and ranking system | Medium | Improves competitiveness |
| 10 | View match history | Medium | Allows users to track past games |
| 11 | Logout and exit system | low | Secondary system control feature |
| 12 | User-friendly interface | High | Improves usability and experience |

# 6. System Design

System design describes the overall structure and working mechanism of the Rock Paper Scissor game .It explains how different modules of the system interact with each other, how data is processed, and how the user interact with the application.

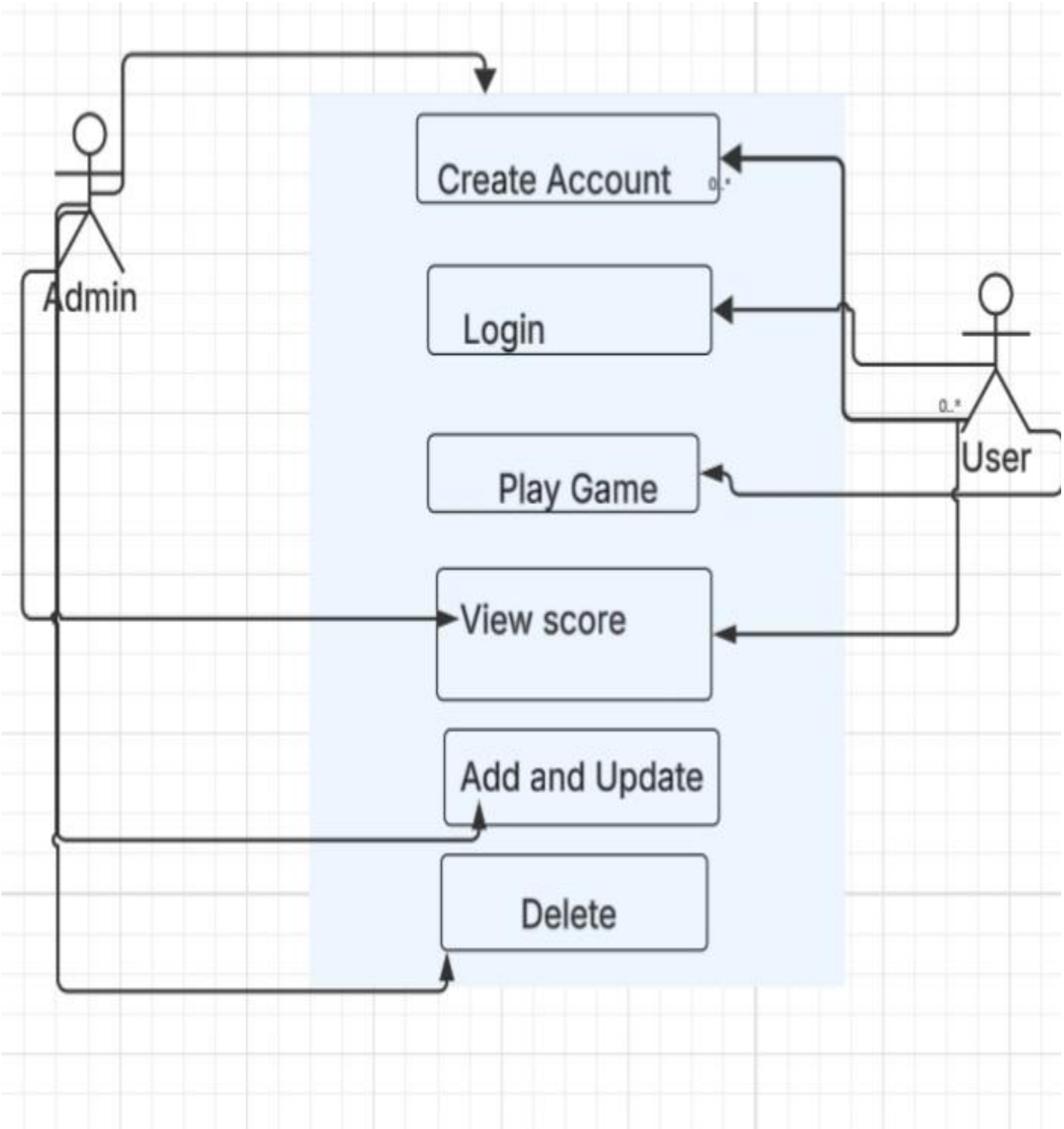


Figure 6. 1

## 7. Detailed Timeline

The waterfall model will be chosen for the development of the Rock Paper Scissor game because it is suitable for small and well defined projects. The project requirements and features will be clearly defined before the development process begins, which will make the waterfall approach effective and easy to manage.

Unlike complex software projects, this project will follow a simple and structured process. Each phase such as requirement analysis, system design, implementation, testing, and maintenance will be completed step by step in sequence. This will help in proper planning, coding, testing, and documentation.

The Waterfall Model will be preferred over other models because:

- The project will have fixed and clearly defined requirements.
- The system will be small-scale and easy to manage.
- Development will be completed phase by phase in an organized manner.
- It will be suitable for academic and documentation-based projects.
- The model will provide clear structure and proper documentation for each stage.

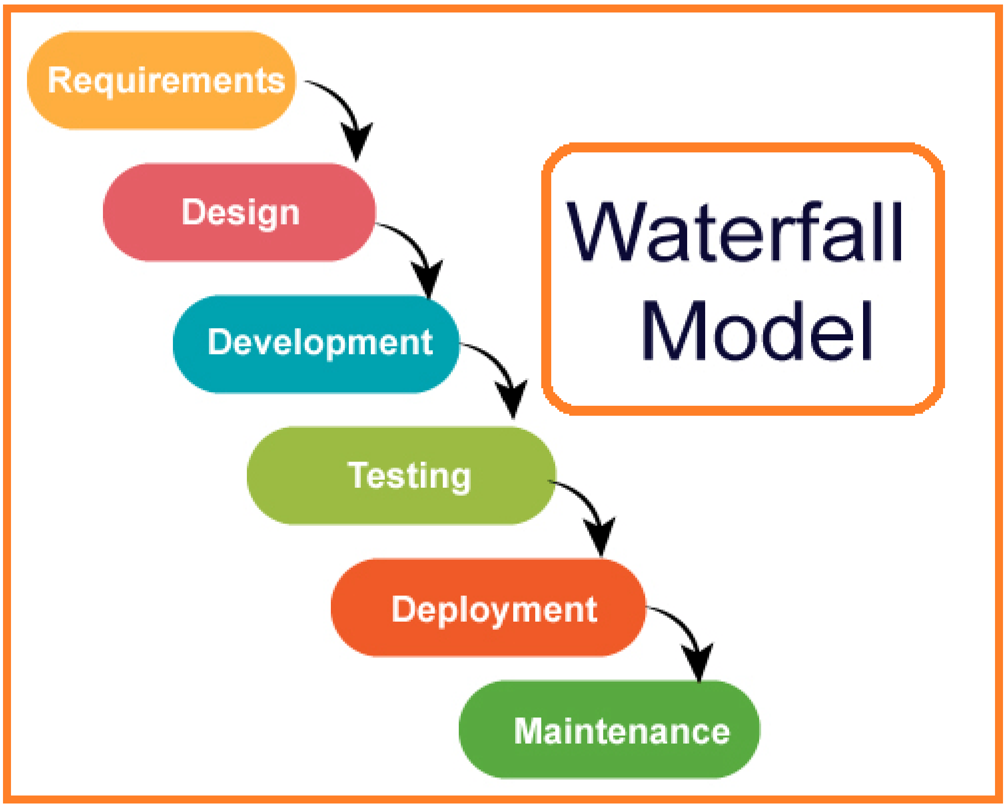


Figure 6. 2

**Gantt chart**

The Gantt chart is a graphical representation of the project schedule that shows the sequence and duration of activities involved in the development of the Rock Paper Scissors game. It helps in planning, monitoring, and completing the project within the allocated time frame.

Since this project follows the Waterfall Model, the development process is divided into different phases such as requirement analysis, system design, coding, testing, documentation, and final deployment. The chart below illustrates the timeline of each phase from the beginning of the project to the final submission period.

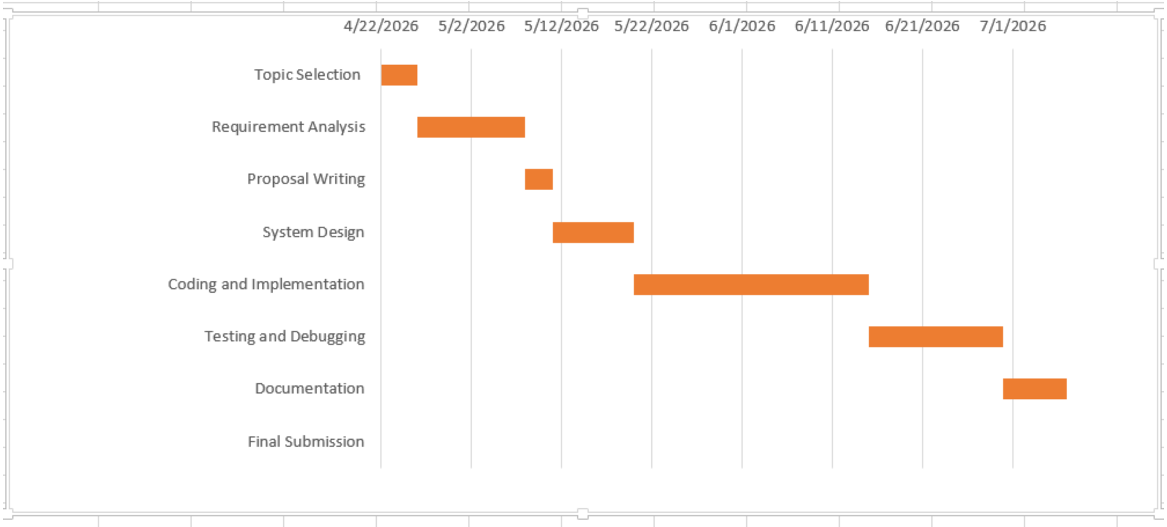


Figure 6. 3

# 8. Conclusion

The Rock Paper Scissors Game project is a console-based desktop application developed using the C programming language. The main purpose of this project is to design and implement a simple, interactive, and user-friendly game system while applying core programming concepts such as file handling, modular programming, and structured problem solving.

This system allows users to register, log in, and play the game either against the computer or another player. It also maintains important records such as match history, scores, and leaderboard rankings. These features make the game more engaging and competitive for users

In addition to the main functionalities, the system will include a dashboard feature in the user interface .This dashboard will act as the main control panel after the user logs in, providing easy access to all available options such as playing the game, viewing match history, checking the leaderboard, and exiting the system. One important function of the dashboard is that it will allow users to view their previous high scores, which helps them track their performance and motivates them to improve their gameplay in future sessions.

Overall, this project will help in understanding real-world software development practices using the Waterfall Model. It will also demonstrate how a structured approach can be used to design, develop, test, and document a complete system in an organized manner.

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